

Physical Chemistry (I)

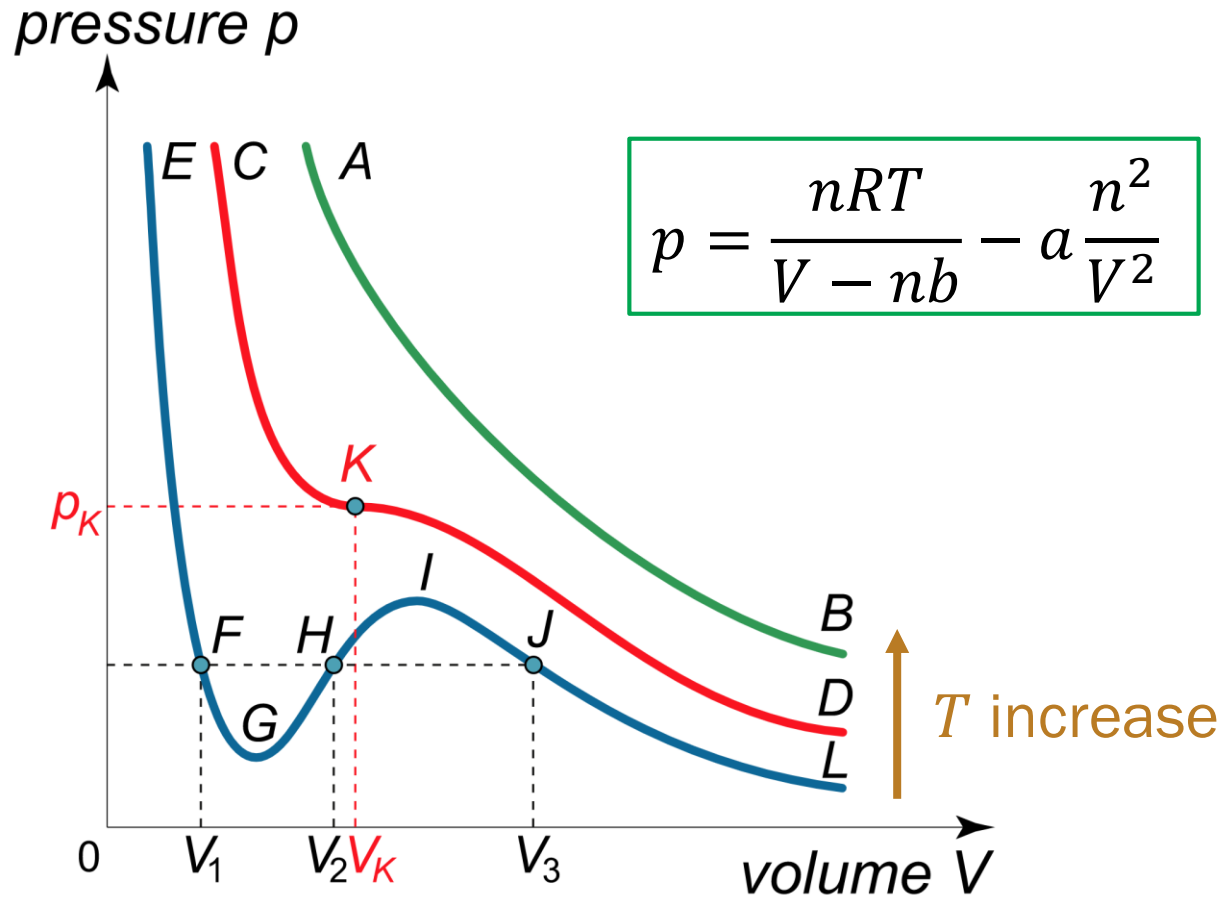
Lecture 14



In the Last Class

- The van der Waals fluid
 - Liquid-gas transition
 - Supercritical fluid
 - Critical point
 - Chemical potentials
- Thermodynamics of phase transitions
 - Phase transitions on (T,p) -space
 - Effect of external pressure on vapor pressure

van der Waals Isotherms



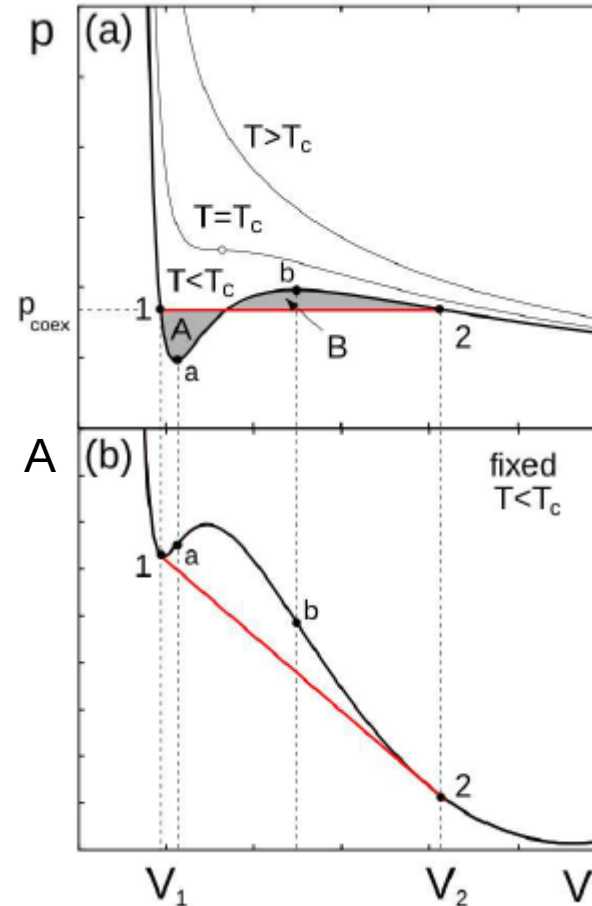
(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

Maxwell Construction

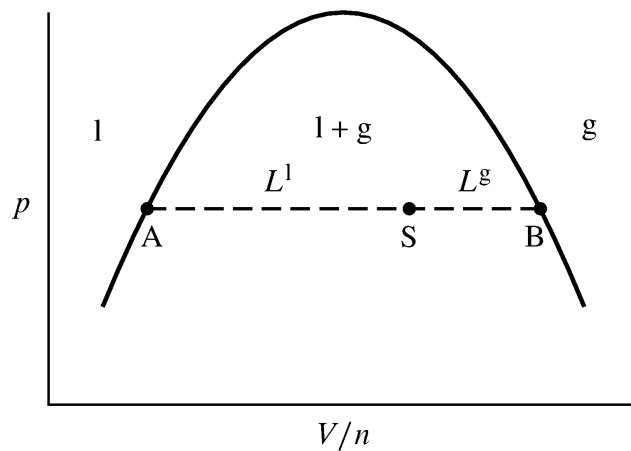
$$p = - \left(\frac{\partial A}{\partial V} \right)_{T,n}$$

At constant T and n ,

$$A = - \int p dV$$

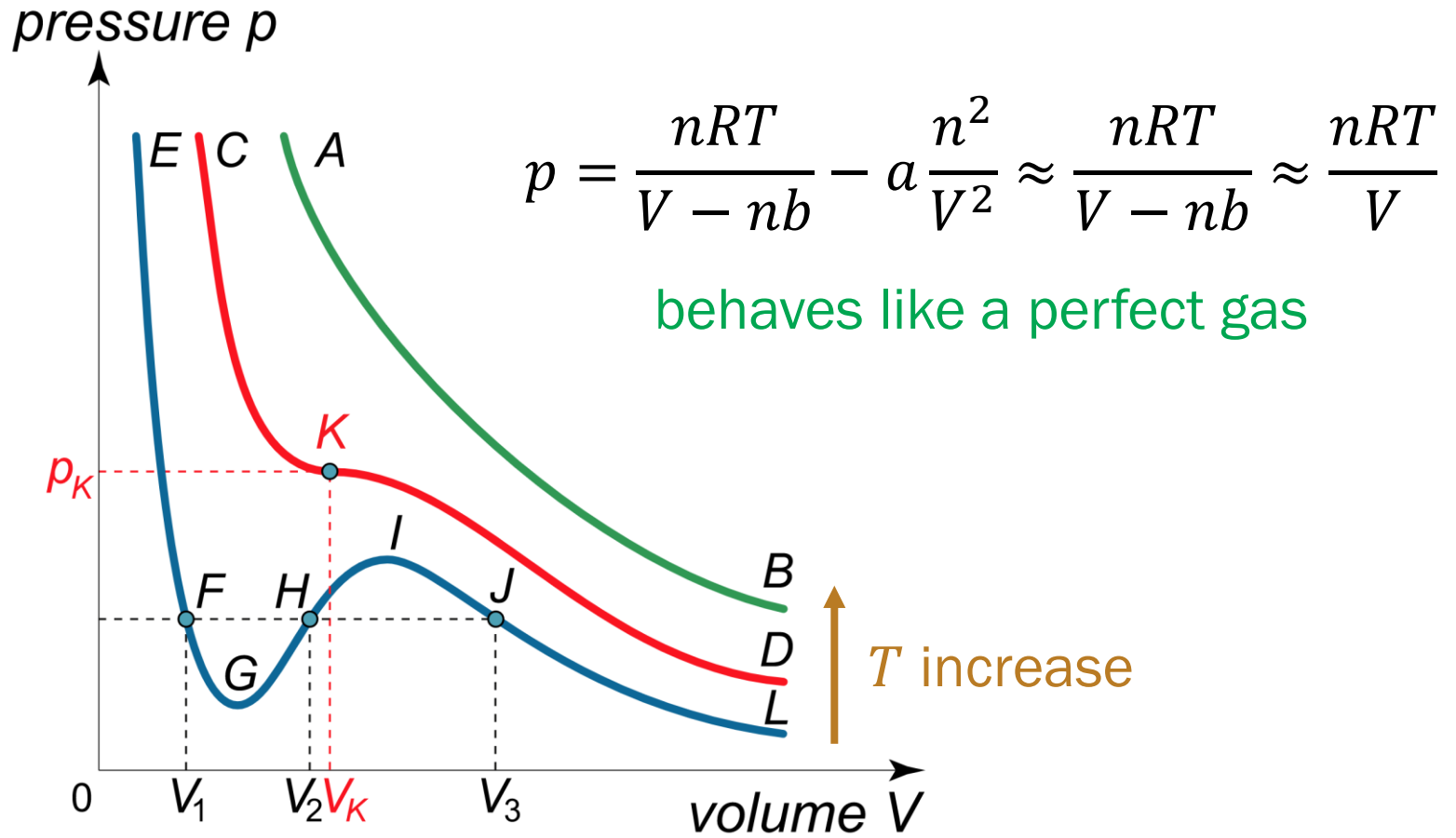


Lever Rule



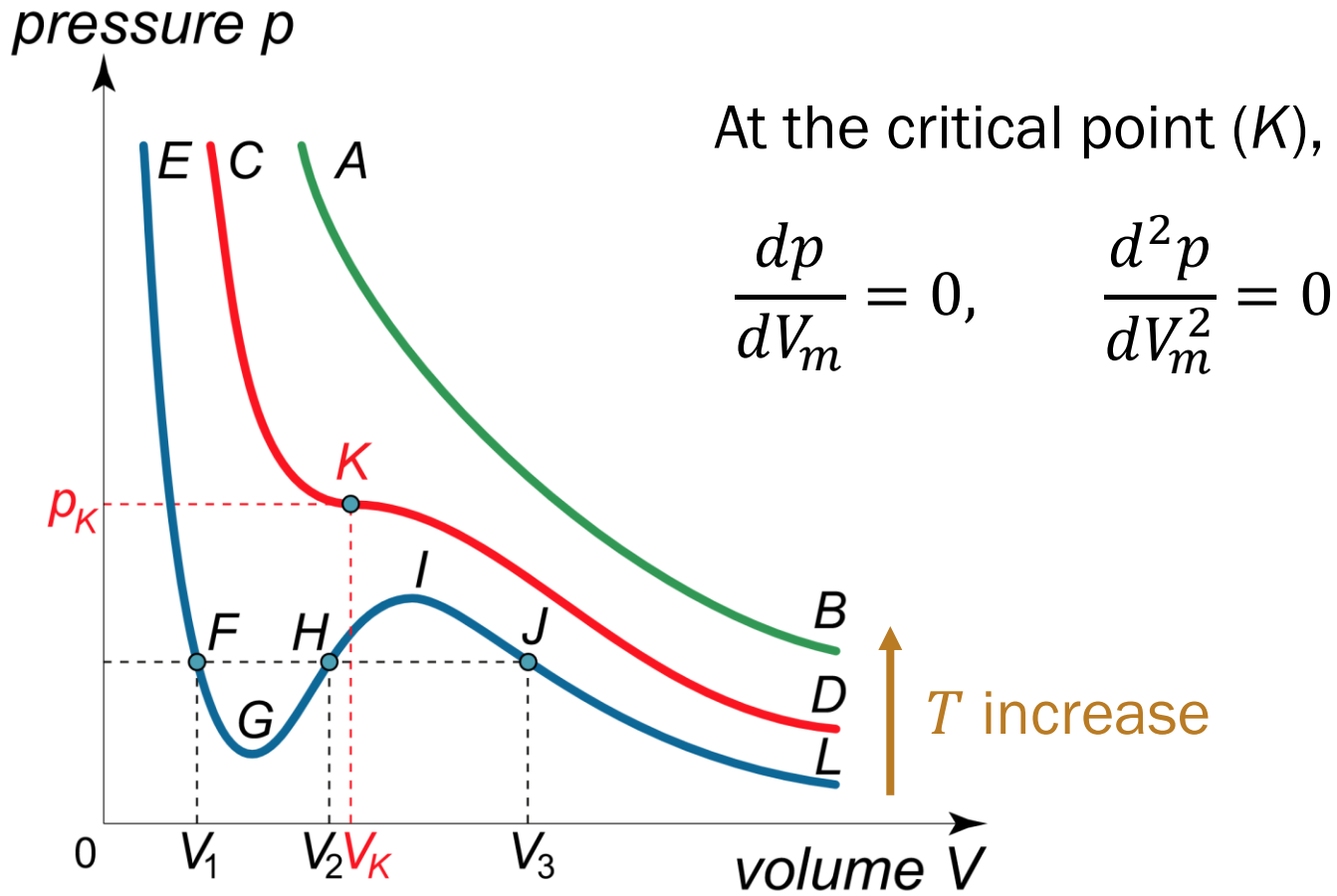
$$n_l : n_g = L^g : L^l$$

Curve A-B: Supercritical Fluid



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

Curve C-D: Critical Point



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

Critical Constants

$$V_c = 3b, \quad p_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27bR}$$

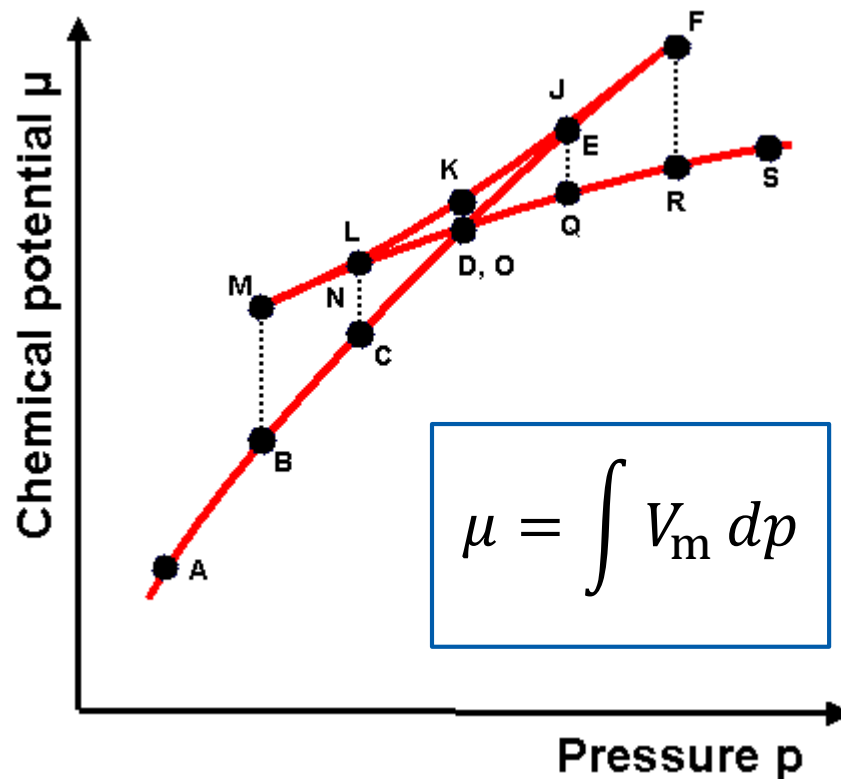
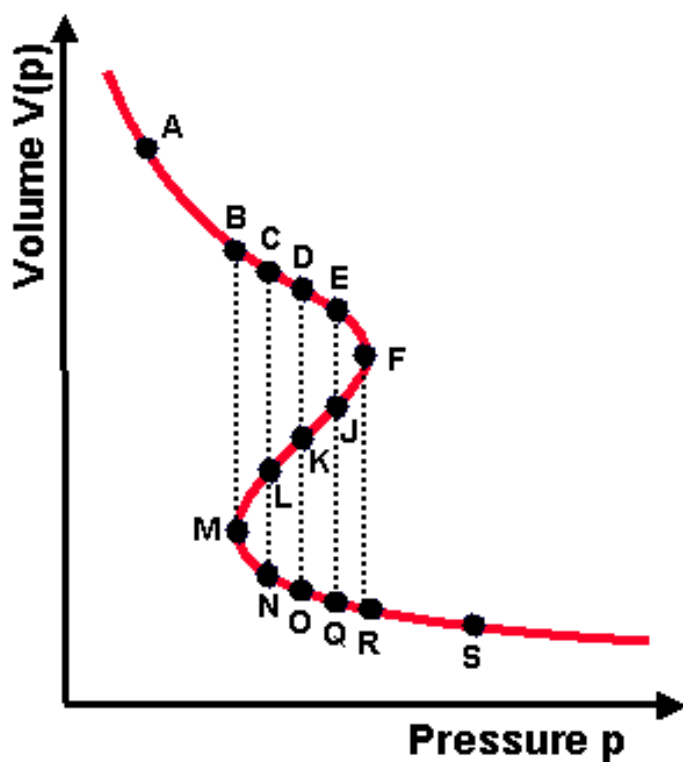
- Critical compression factor

$$Z_c = \frac{p_c V_c}{RT_c} = \frac{3}{8}$$

- Reduced variables: $p_r = p/p_c$, $V_r = V_m/V_c$, $T_r = T/T_c$

$$p_r = \frac{8T_r}{3V_r - 1} - \frac{3}{V_r^2}$$

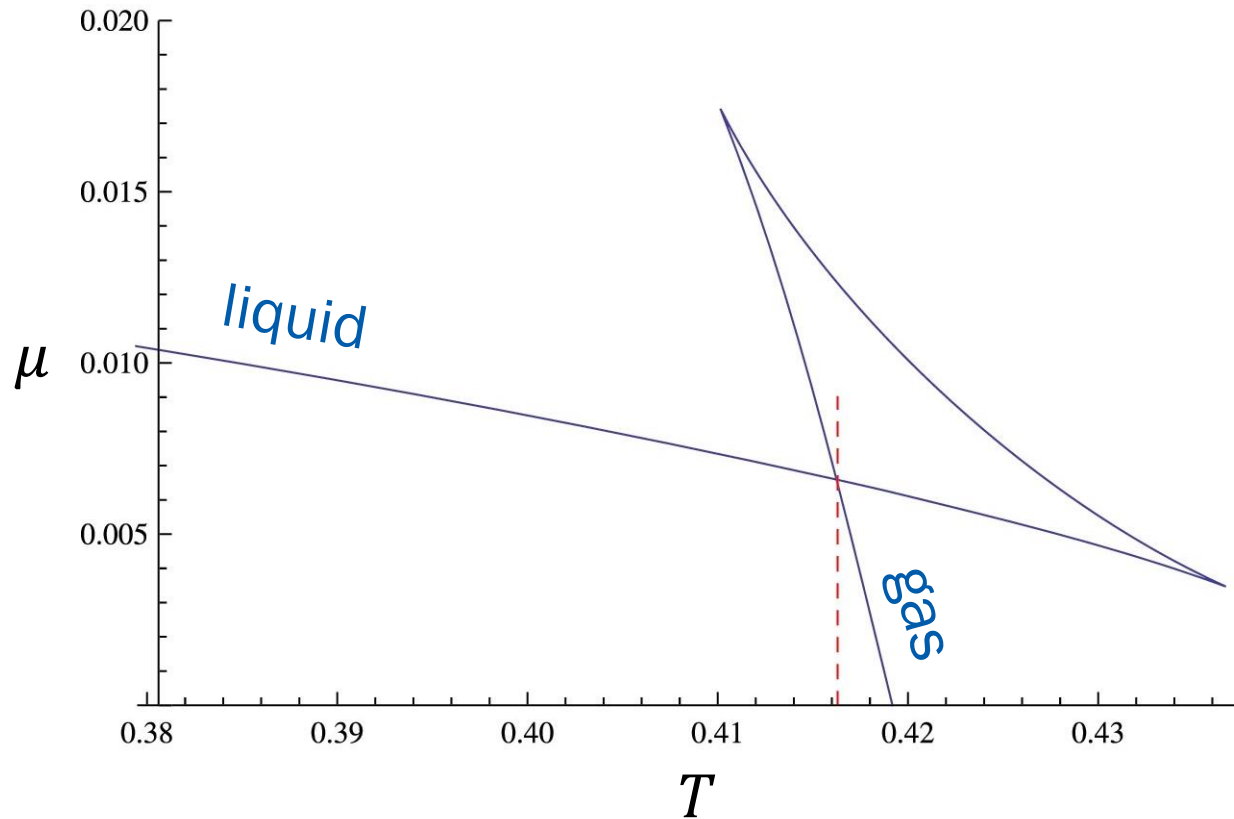
Chemical Potential $\mu(p)$



$$\mu = \int V_m dp$$

(Image: http://casey.brown.edu/chemistry/misspelled-research/crp/Edu/Documents/00_Chem201/7_phase_equil_1/Chem201_7_phase_equil_1.htm)

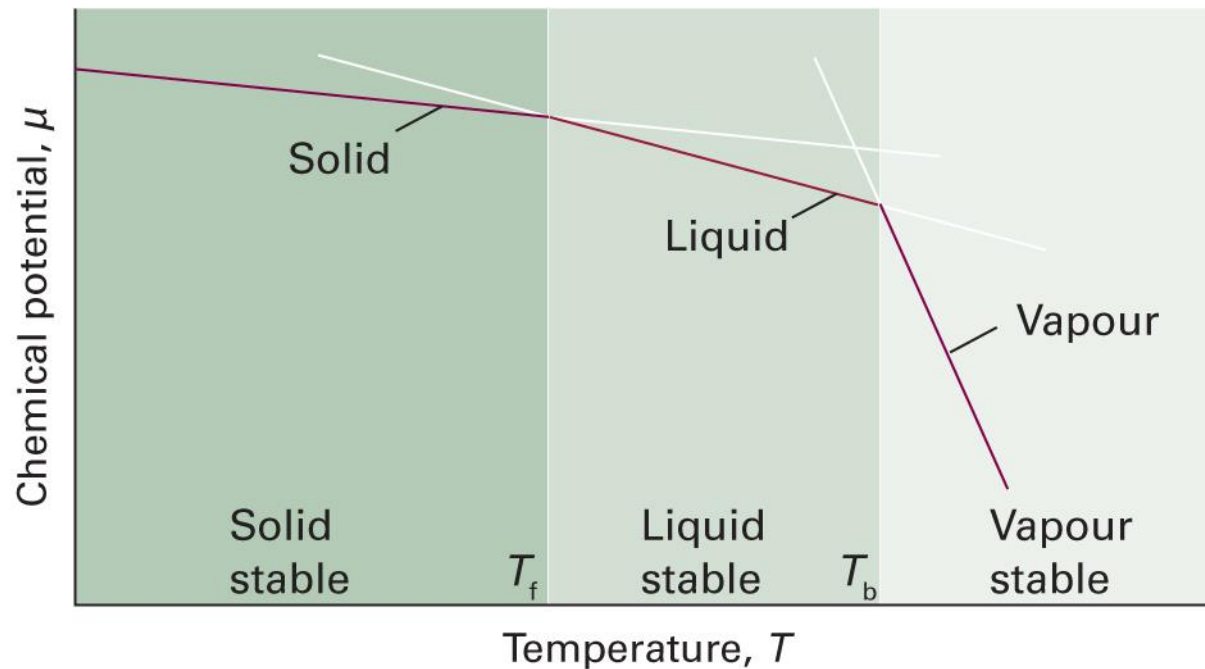
Chemical Potential $\mu(T)$



(Image: He et al., *Nuclear Physics B* 915: 243-261 (2017), Figure 3)

Chemical Potential and T

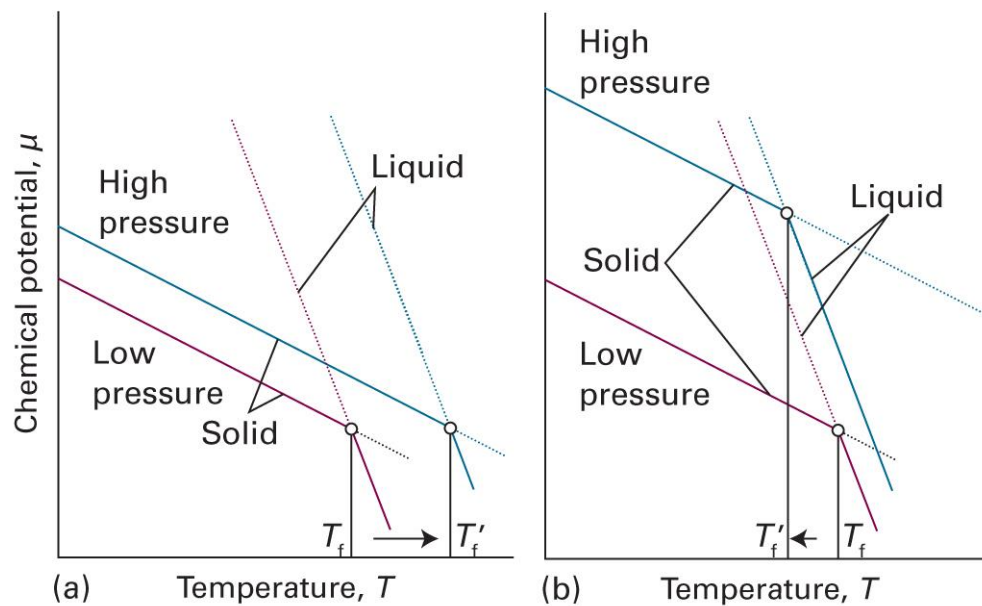
$$\left(\frac{\partial \mu}{\partial T}\right)_{p,n} = -S_m$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.1)

Chemical Potential and p

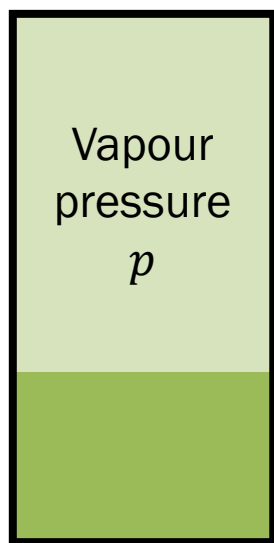
$$\left(\frac{\partial \mu}{\partial p}\right)_{T,n} = V_m$$



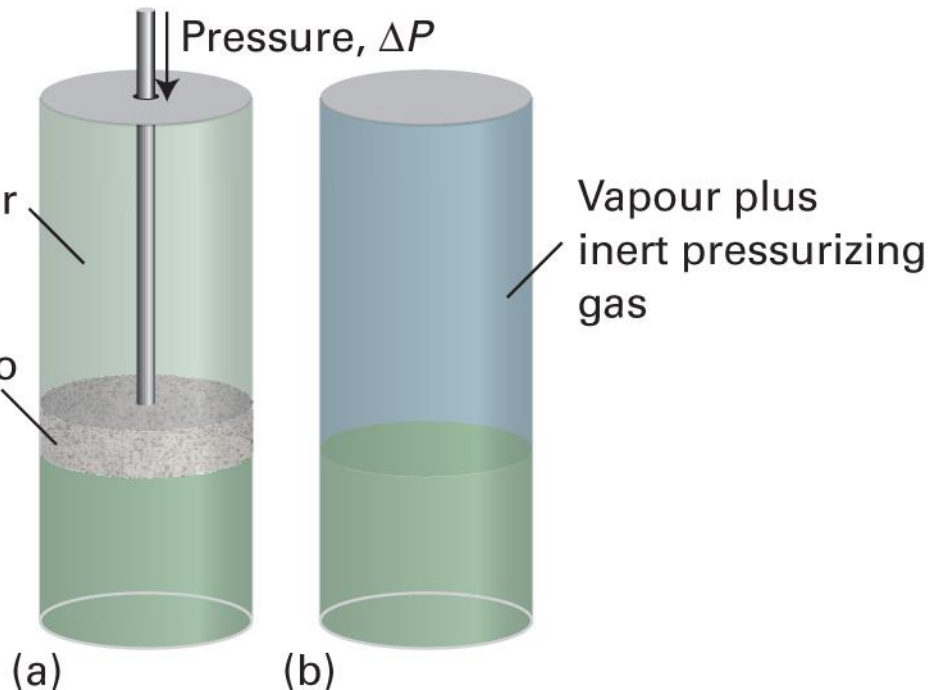
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.2)

Pressure on Liquid-Gas Eq.

Liquid + Gas



Piston permeable to vapour but not liquid



$$p = p^* e^{V_m(l)\Delta P/RT}$$

Phase Boundaries

When two phases α and β are at equilibrium,

$$\mu(\alpha; p, T) = \mu(\beta; p, T)$$

For infinitesimal changes dp and dT ,

$$d\mu(\alpha) = d\mu(\beta)$$

Since $d\mu = V_m dp - S_m dT$ (Gibbs-Duhem equation),

$$V_m(\alpha)dp - S_m(\alpha)dT = V_m(\beta)dp - S_m(\beta)dT$$

Clapeyron Equation

$$V_m(\alpha)dp - S_m(\alpha)dT = V_m(\beta)dp - S_m(\beta)dT$$

Rearrangement leads to

$$\{S_m(\beta) - S_m(\alpha)\}dT = \{V_m(\beta) - V_m(\alpha)\}dp$$

$$\Delta_{\text{trs}}SdT = \Delta_{\text{trs}}Vdp$$

Thus,

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}S}{\Delta_{\text{trs}}V}, \quad \frac{dT}{dp} = \frac{\Delta_{\text{trs}}V}{\Delta_{\text{trs}}S}$$

Brief Illustration 4B.3

For water at 0°C, the standard volume of transition of ice to liquid is $-1.6 \text{ cm}^3 \text{ mol}^{-1}$, and the corresponding standard entropy of transition is $+22 \text{ J K}^{-1} \text{ mol}^{-1}$. What is the slope of the solid–liquid phase boundary at that temperature?

$$\begin{aligned} dT/dp &= \Delta_{\text{trs}}V/\Delta_{\text{trs}}S = (-1.6 \text{ cm}^3 \text{ mol}^{-1}) \times (10^{-6} \text{ m}^3 \text{ cm}^{-3}) / (22 \text{ J K}^{-1} \text{ mol}^{-1}) \\ &= -7.3 \times 10^{-8} \text{ K Pa}^{-1} \end{aligned}$$

(Almost Always) Positive Slope

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}S}{\Delta_{\text{trs}}V}$$

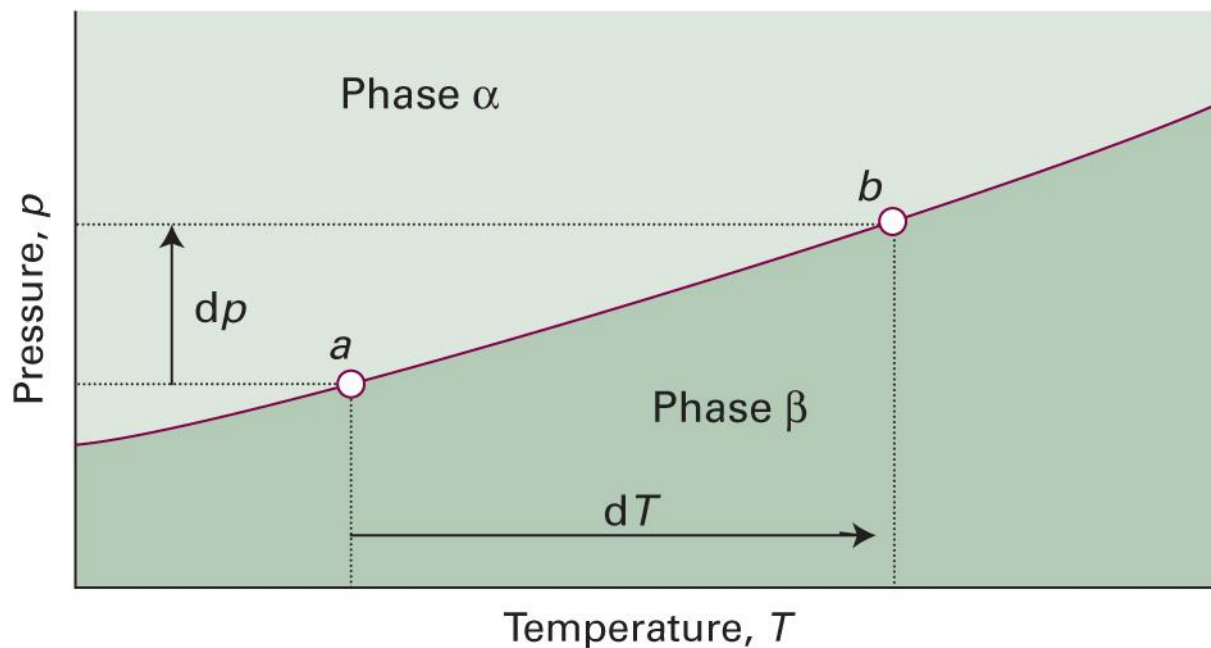
For a phase transition, $\Delta_{\text{trs}}S = \Delta_{\text{trs}}H/T$, and

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}H}{T\Delta_{\text{trs}}V}$$

Hence, dp/dT is almost always positive (unless $\Delta_{\text{trs}}V < 0$)

Phase Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}H}{T\Delta_{\text{trs}}V}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.4)

Solid-Liquid Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{fus}}H}{T \Delta_{\text{fus}}V}$$

Assuming $\Delta_{\text{fus}}H$ and $\Delta_{\text{fus}}V$ are constant,

$$dp = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \frac{1}{T} dT$$

$$\int_{p_1}^{p_2} dp = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \int_{T_1}^{T_2} \frac{1}{T} dT$$

Solid-Liquid Boundary

$$\int_{p_1}^{p_2} dp = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \int_{T_1}^{T_2} \frac{1}{T} dT$$

After integration,

$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \ln \left(\frac{T_2}{T_1} \right)$$

Solid-Liquid Boundary

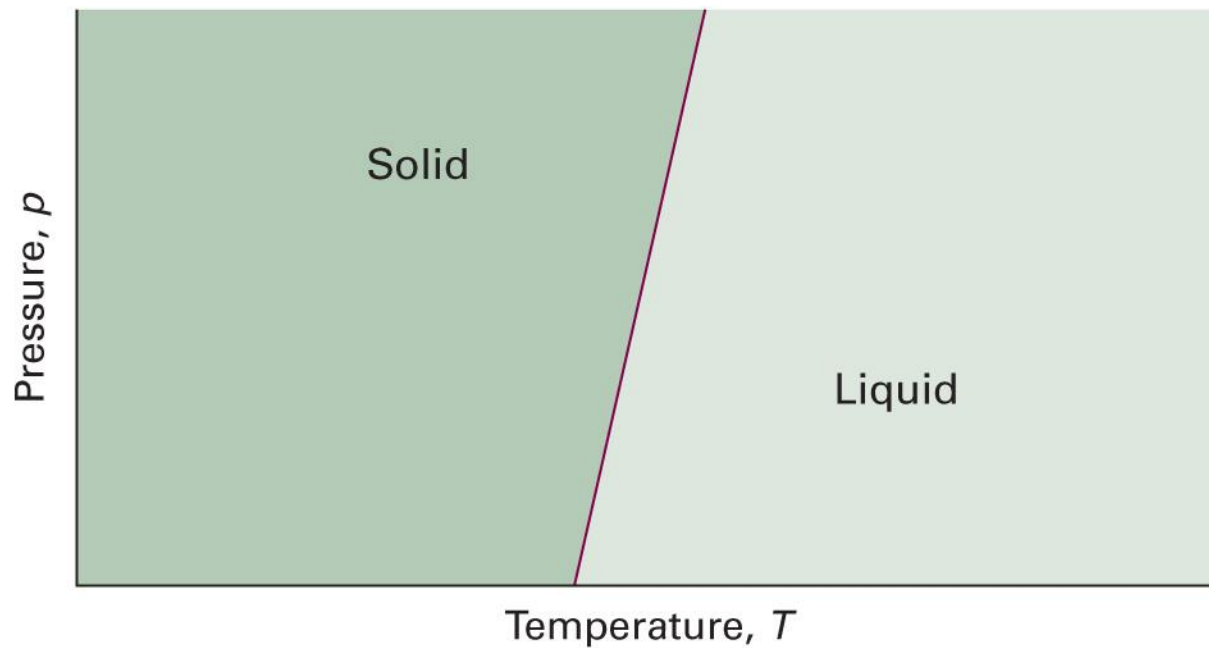
$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \ln \left(\frac{T_2}{T_1} \right)$$

If $T_1 \approx T_2$, using $\ln(1 + x) = x - x^2/2 + \dots$,

$$\ln \left(\frac{T_2}{T_1} \right) = \ln \left(1 + \frac{T_2 - T_1}{T_1} \right) \approx \frac{T_2 - T_1}{T_1}$$

$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{T_1 \Delta_{\text{fus}}V} (T_2 - T_1)$$

Solid-Liquid Boundary



Liquid-Gas Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{vap}}H}{T\Delta_{\text{vap}}V}$$

Since $V_{\text{m}}(g) \gg V_{\text{m}}(l)$,

$$\Delta_{\text{vap}}V = V_{\text{m}}(g) - V_{\text{m}}(l) \approx V_{\text{m}}(g) \approx RT/p$$

Thus,

$$\frac{1}{p} \frac{dp}{dT} = \frac{\Delta_{\text{vap}}H}{RT^2}$$

(Clausius-Clapeyron equation)

Liquid-Gas Boundary

$$\frac{1}{p} dp = \frac{\Delta_{\text{vap}}H}{RT^2} dT$$

Assuming $\Delta_{\text{vap}}H$ is constant,

$$\int_{p_1}^{p_2} \frac{1}{p} dp = \frac{\Delta_{\text{vap}}H}{R} \int_{T_1}^{T_2} \frac{1}{T^2} dT$$

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Liquid-Gas Boundary

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

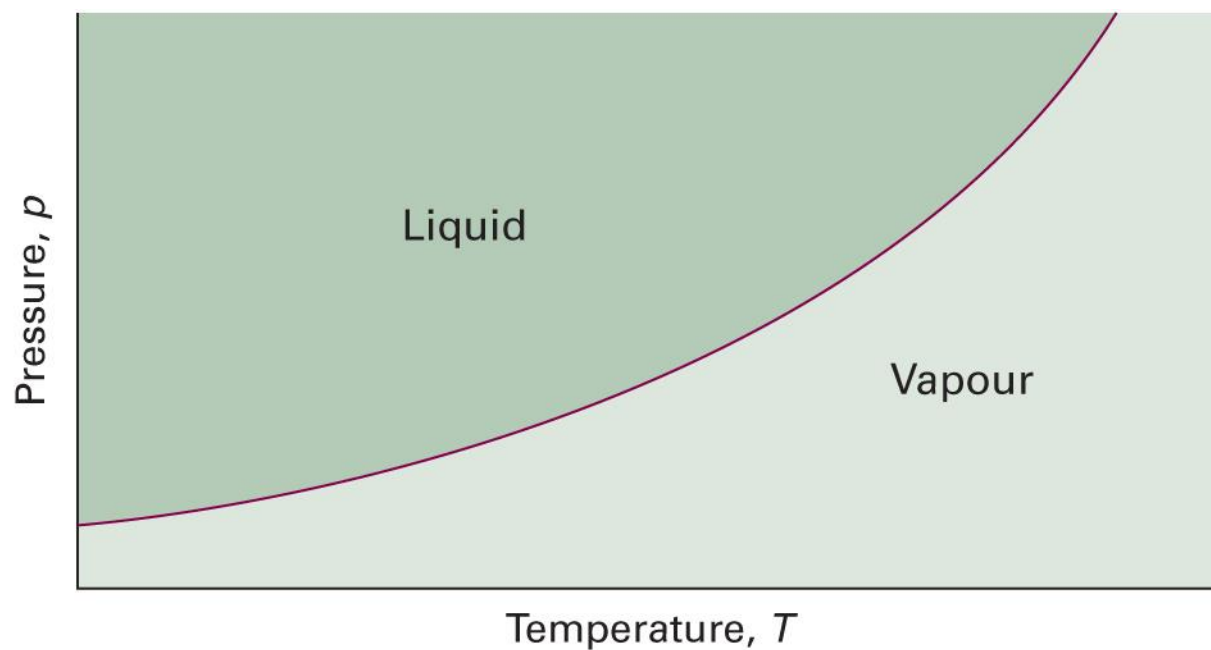
Hence,

$$p_2 = p_1 e^{A-B/T_2}$$

where

$$A = \frac{\Delta_{\text{vap}}H}{RT_1}, \quad B = \frac{\Delta_{\text{vap}}H}{R}$$

Liquid-Gas Boundary



Brief Illustration 4B.5

The normal boiling point of benzene is 80°C (353 K) and $\Delta_{\text{vap}}H^\ominus = 30.8 \text{ kJ mol}^{-1}$. Calculate the vapour pressure at 20°C (293 K).

$$\ln\left(\frac{p_2}{p_1}\right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Let $p_1 = 1.00 \text{ atm}$, $p_2 = p$, $T_1 = 353 \text{ K}$, and $T_2 = 293 \text{ K}$.

The right-hand side of the equation is

$$(3.08 \times 10^4 \text{ J mol}^{-1}) / (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (1/353 \text{ K} - 1/293 \text{ K}) = -2.14$$

Hence,

$$p = (1.00 \text{ atm})e^{-2.14} = 0.118 \text{ atm}$$

Solid-Gas Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{sub}}H}{T\Delta_{\text{sub}}V}$$

Similarly, $\Delta_{\text{sub}}V \approx V_{\text{m}}(g) \approx RT/p$, and

$$\frac{1}{p} \frac{dp}{dT} = \frac{\Delta_{\text{sub}}H}{RT^2}$$

Assuming $\Delta_{\text{sub}}H$ is constant,

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta_{\text{sub}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

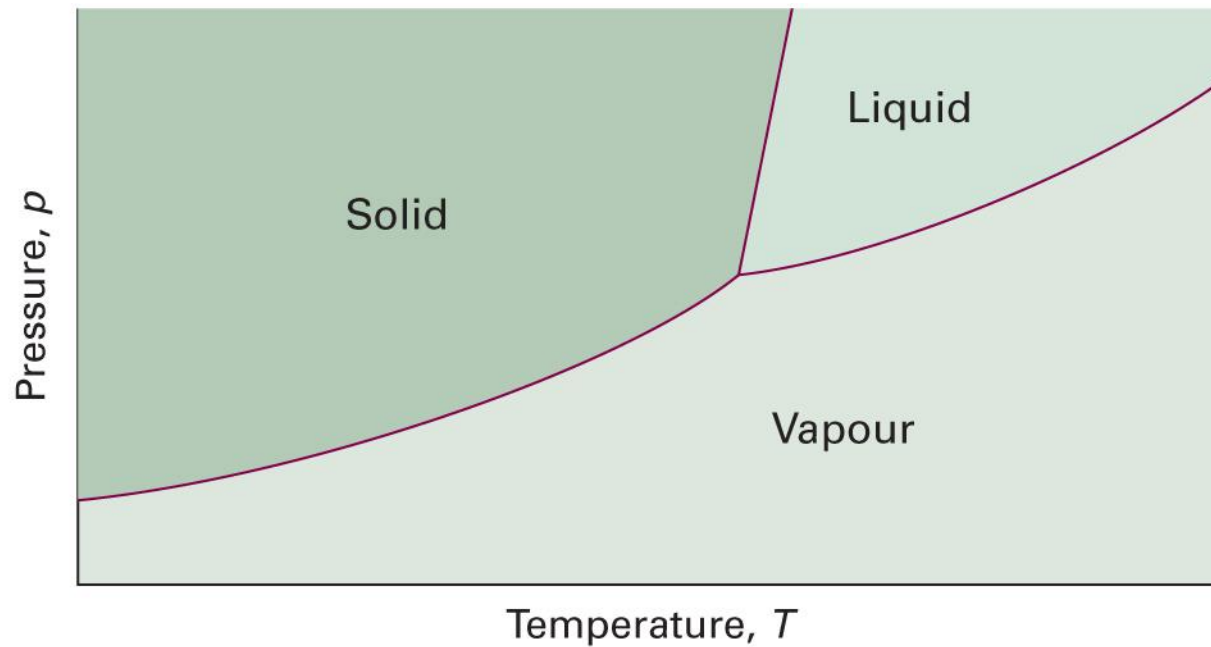
Solid-Gas vs. Liquid-Gas

$$\text{Solid-gas: } \ln\left(\frac{p_2}{p_1}\right) = \frac{\Delta_{\text{sub}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\text{Liquid-gas: } \ln\left(\frac{p_2}{p_1}\right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

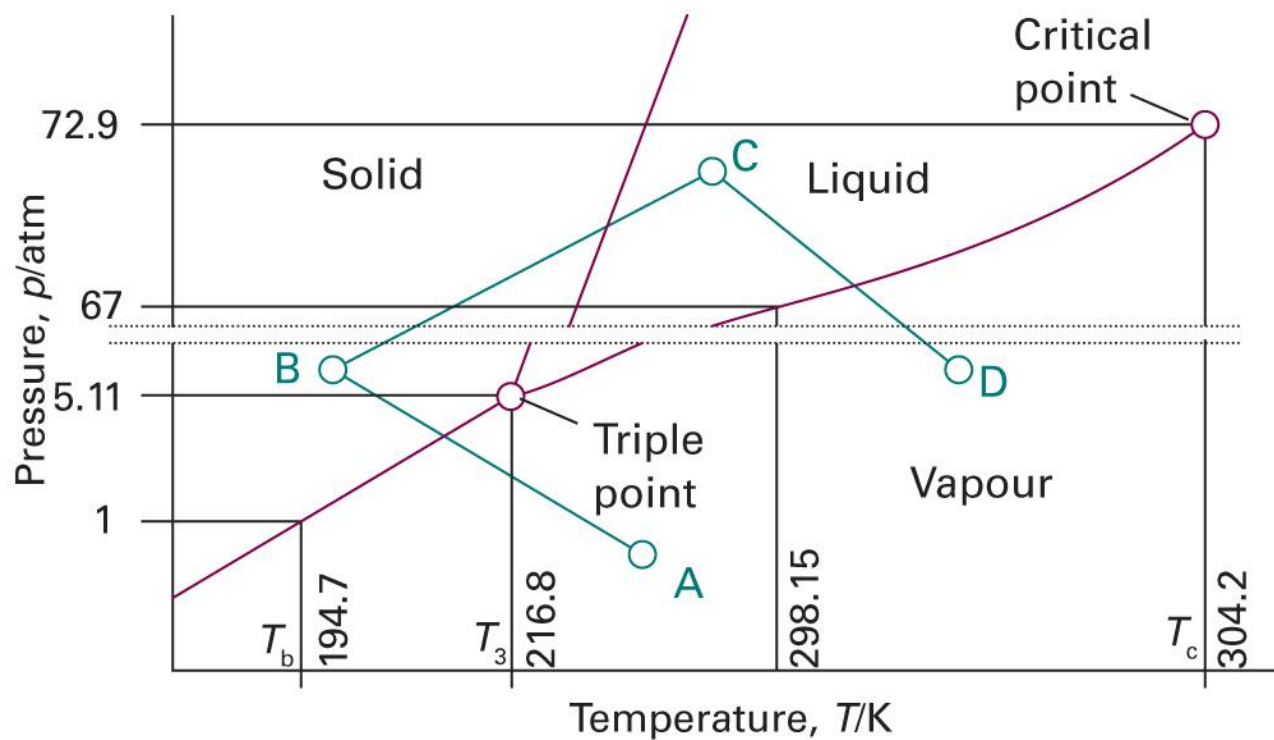
Since $\Delta_{\text{sub}}H = \Delta_{\text{fus}}H + \Delta_{\text{vap}}H > \Delta_{\text{vap}}H$, the **sublimation curve** has a **steeper slope** than the **vaporization curve**

Phase Diagram



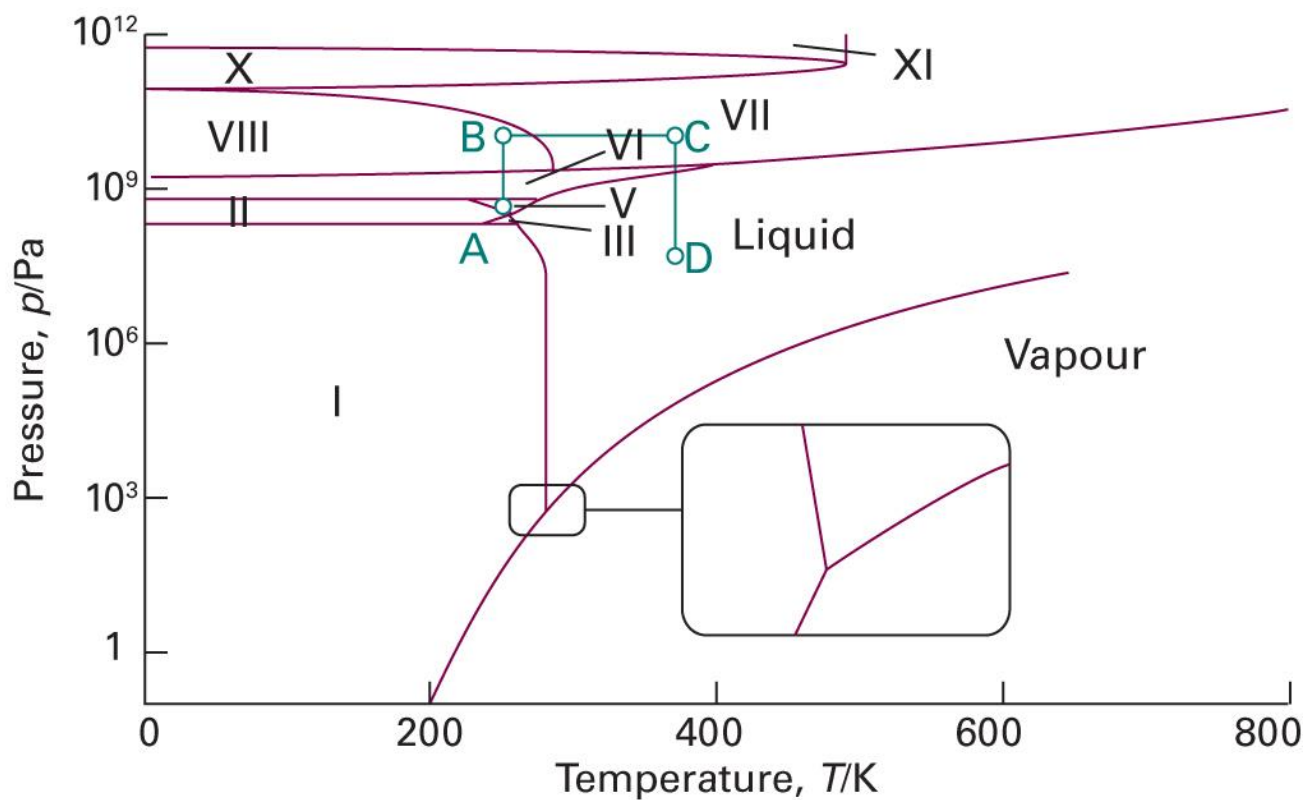
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.7)

Revisiting CO₂



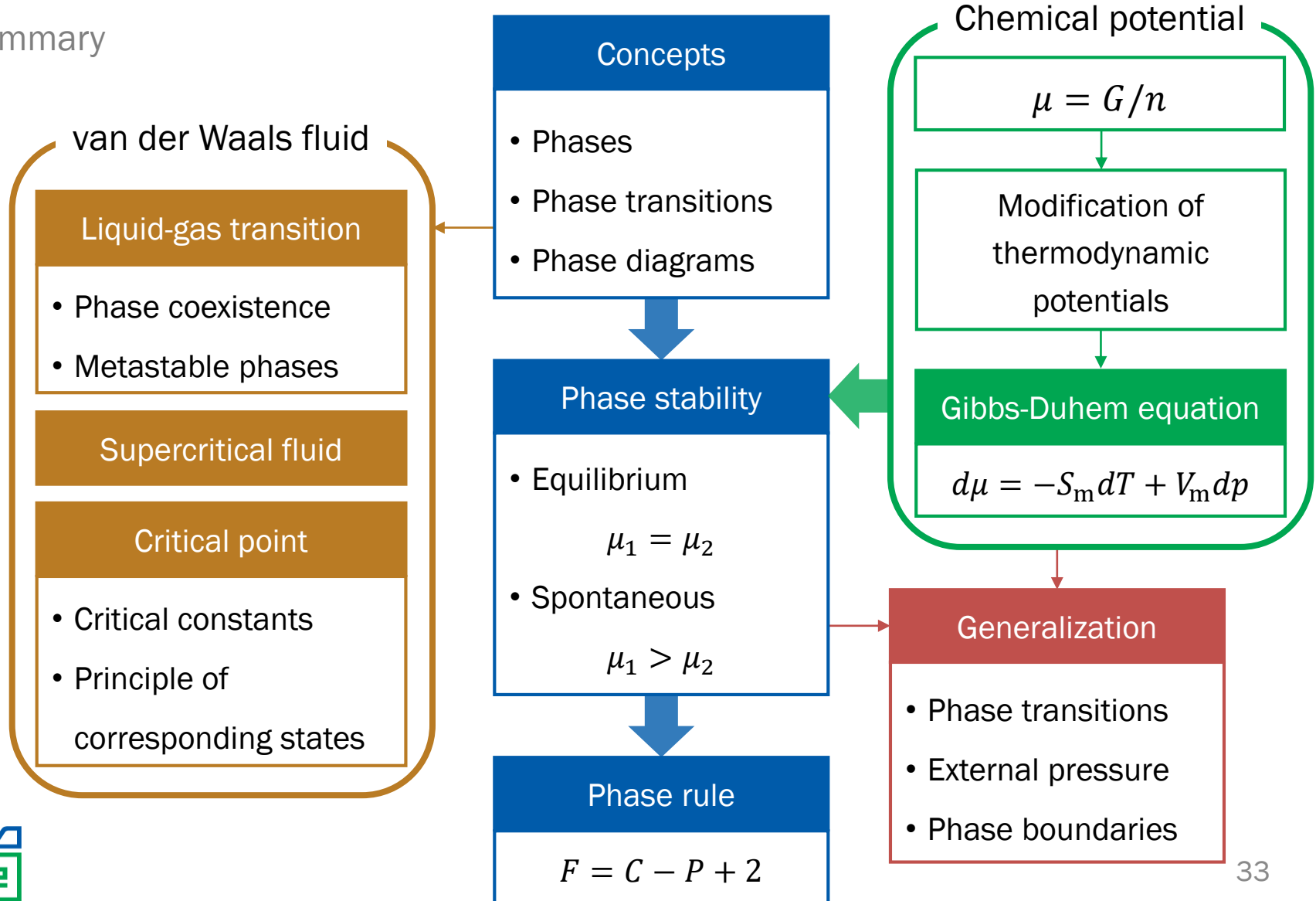
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.8)

Revisiting H₂O



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.9)

Summary



7. Simple Mixtures

Chapter Overview

- Multi-component systems
- Mixing of gases
 - Mixing of perfect gases
 - Mixing of real gases
- Mixing of liquids and solids
 - Ideal solutions
 - Real solutions
 - Liquid-solid solutions: colligative properties
- Ternary systems



Multi-Component System

- State: $(p, T, n) \rightarrow (p, T, n_A, n_B, \dots)$
- Thermodynamic potentials
 - $dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$

Multi-Component System

- Possible phases
 - Each component: gas, liquid, solid
 - Mixed/demixed
- e.g. two-component system (binary system)
 - A: liquid, B: liquid, A-B: mixed
 - A: liquid, B: liquid, A-B: demixed
 - A: liquid, B: gas, A-B: demixed
 - ...

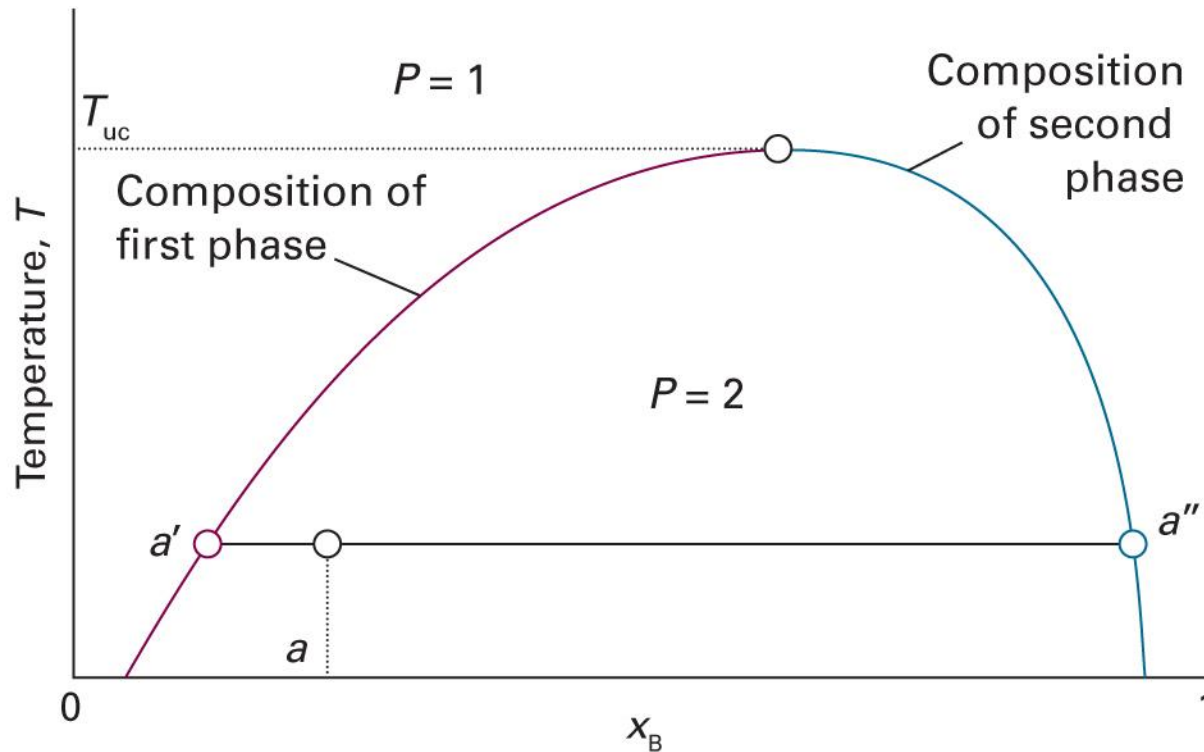
Phase Diagram

- Mole fraction of species J

$$x_J = \frac{n_J}{\sum_J n_J}$$

- At least one axis (usually the x-axis) is for x_J
- The other is for $T, p, \dots$

Example: Binary System



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.13)

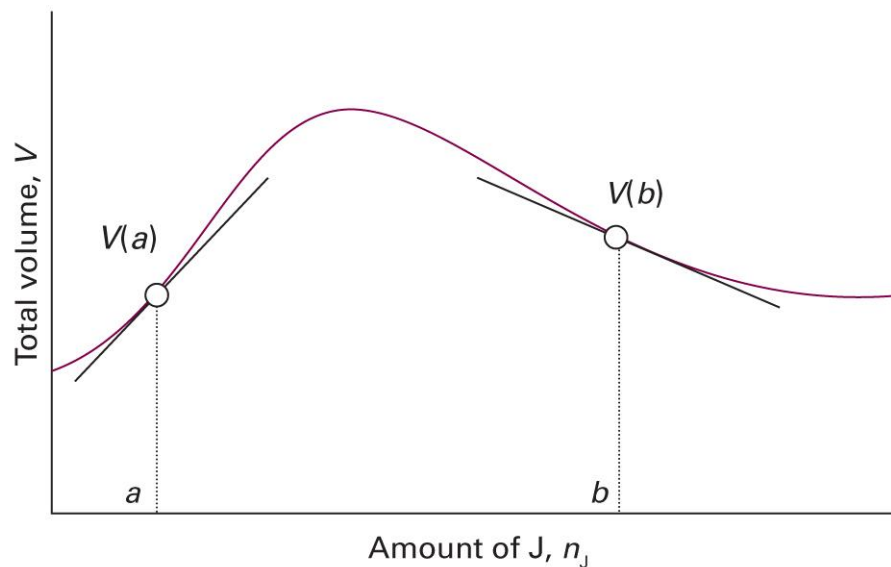
Partial Molar Quantities

$$X_J = \left(\frac{\partial X}{\partial n_J} \right)_{p,T,n'}$$

contribution that a component of a mixture makes
to the total quantity of a sample

Partial Molar Volume

$$V_J = \left(\frac{\partial V}{\partial n_J} \right)_{p,T,n'}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.2)

Change of Volume

For a two-component system at constant p and T ,

$$V = V(n_A, n_B)$$

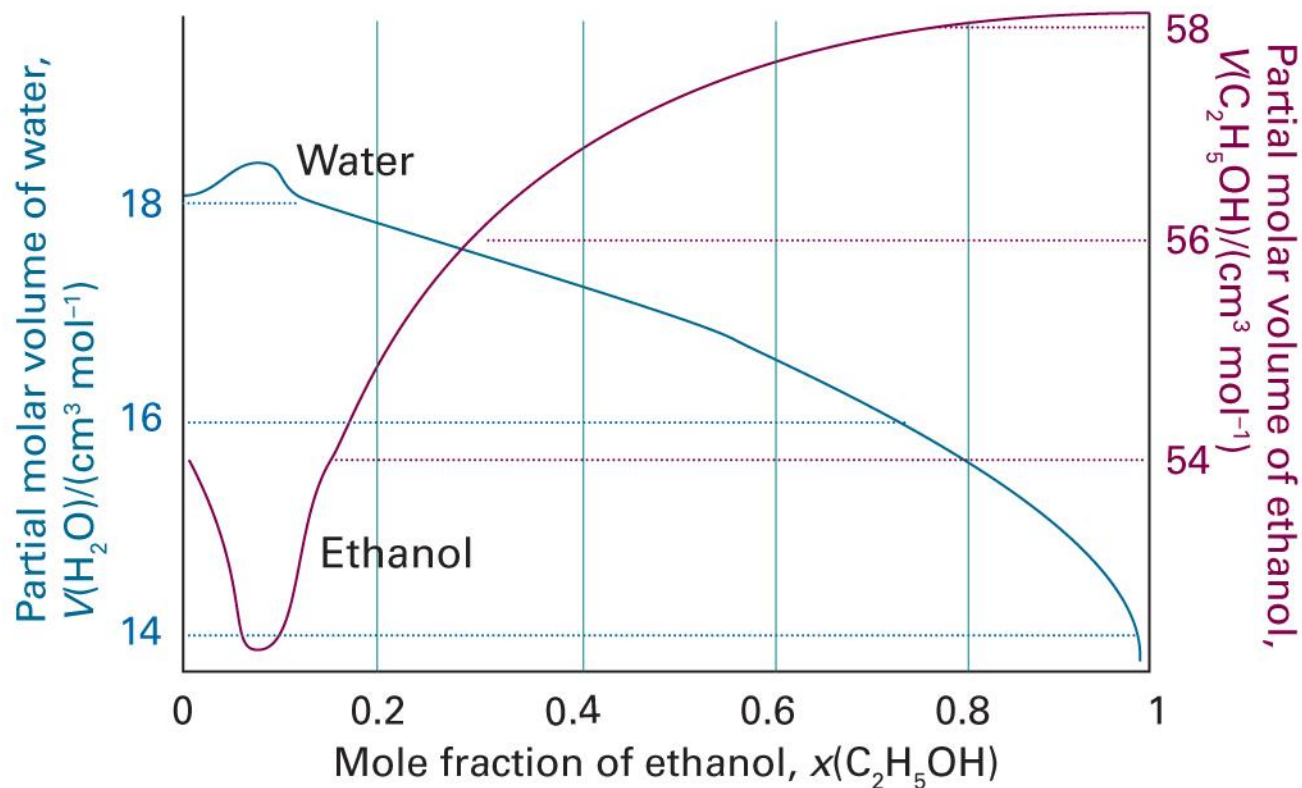
leads to

$$dV = \left(\frac{\partial V}{\partial n_A} \right)_{p,T,n_B} dn_A + \left(\frac{\partial V}{\partial n_B} \right)_{p,T,n_A} dn_B$$

Hence,

$$dV = V_A dn_A + V_B dn_B$$

Example: Water + EtOH



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.1)

Example 5A.1

A polynomial fit to measurements of the total volume of a water/ethanol mixture at 25°C that contains 1.000 kg of water is

$$v = 1002.93 + 54.6664z - 0.36394z^2 + 0.028256z^3$$

where $v = V/\text{cm}^3$, $z = n_{\text{E}}/\text{mol}$, and n_{E} is the amount of EtOH present. Determine the partial molar volume of ethanol.

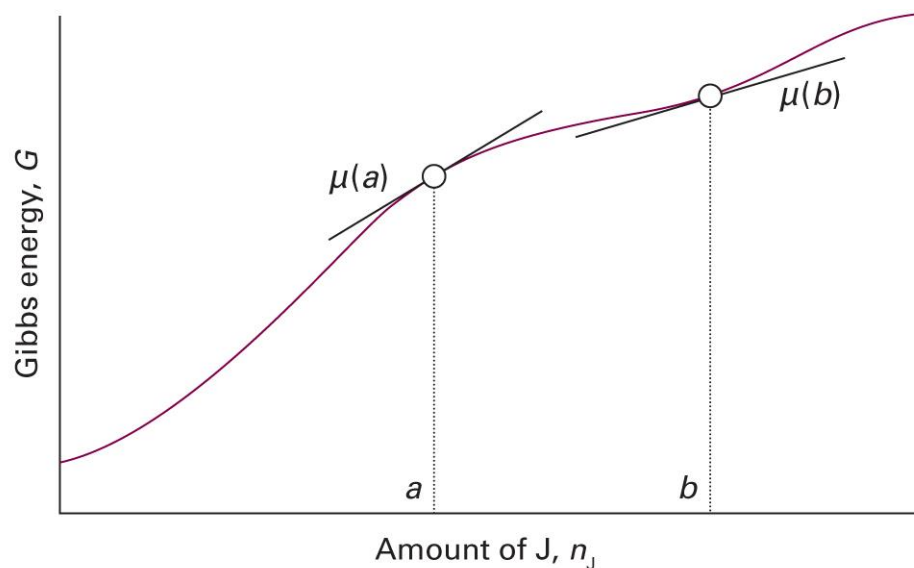
$$V_{\text{E}} = (\partial V / \partial n_{\text{E}})_{p,T,n_{\text{W}}} = (\partial v / \partial z)_{p,T,n_{\text{W}}} \text{ cm}^3 \text{ mol}^{-1}$$

Since $(\partial v / \partial z)_{p,T,n_{\text{W}}} = 54.6664 - 0.72788z + 0.084768z^2$,

$$V_{\text{E}} = (54.6664 - 0.72788z + 0.084768z^2) \text{ cm}^3 \text{ mol}^{-1}$$

Partial Gibbs Energy

$$\mu_J = \left(\frac{\partial G}{\partial n_J} \right)_{p,T,n'}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.4)

... is Chemical Potential

$$dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$$

Hence,

$$\mu_J = \left(\frac{\partial U}{\partial n_J} \right)_{S,V,n'} = \left(\frac{\partial H}{\partial n_J} \right)_{S,p,n'} = \left(\frac{\partial A}{\partial n_J} \right)_{T,V,n'} = \left(\frac{\partial G}{\partial n_J} \right)_{T,p,n'}$$

Not in Atkins

Be Careful

$$\mu_J = \left(\frac{\partial U}{\partial n_J} \right)_{S,V,n'} = \left(\frac{\partial H}{\partial n_J} \right)_{S,p,n'} = \left(\frac{\partial A}{\partial n_J} \right)_{T,V,n'} = \left(\frac{\partial G}{\partial n_J} \right)_{T,p,n'}$$

$$U_J = \left(\frac{\partial U}{\partial n_J} \right)_{T,p,n'} \neq \mu_J$$

$$H_J = \left(\frac{\partial H}{\partial n_J} \right)_{T,p,n'} \neq \mu_J$$

$$A_J = \left(\frac{\partial A}{\partial n_J} \right)_{T,p,n'} \neq \mu_J$$

Gibbs-Duhem Equation

Now, thermodynamic equations for internal energy are

$$U = TS - pV + \sum_J \mu_J n_J$$

$$dU = TdS - pdV + \sum_J \mu_J dn_J$$

Differentiating the first equation,

$$dU = TdS + SdT - pdV - Vdp + \sum_J \mu_J dn_J + \sum_J n_J d\mu_J$$

Comparison leads to

$$SdT - Vdp + \sum_J n_J d\mu_J = 0$$

At Constant T and p

$$SdT - Vdp + \sum_J n_J d\mu_J = 0$$

At constant T and p ,

$$\sum_J n_J d\mu_J = 0$$

For a two-component system,

$$n_A d\mu_A + n_B d\mu_B = 0$$

Brief Illustration 5A.1

If the composition of a mixture is such that $n_A = 2n_B$, and a small change in composition results in μ_A changing by $\Delta\mu_A = +1 \text{ J mol}^{-1}$, what is the change in μ_B ?

Since $n_A d\mu_A + n_B d\mu_B = 0$,

$$\Delta\mu_B \approx -(n_A/n_B)\Delta\mu_A = -2 \times (+1 \text{ J mol}^{-1}) = -2 \text{ J mol}^{-1}$$